

The Honest-Reporting Gap in Quantum Error Correction: A Tamper-Evident Receipt Schema for Acceptance-Fraction Disclosure

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Abstract—Reported quantum error correction (QEC) gains routinely omit a single critical quantity: the fraction of shots that were accepted rather than discarded via post-selection (selective rejection). The landmark Microsoft/Quantinuum *Nature* result (2026) reports an 800-fold logical-vs-physical error suppression using the carbon code on trapped-ion hardware—a number derived from selectively rejected runs. The non-rejecting (filtered) figure is approximately 5-fold; the mid-computation figure, which is the one fault-tolerant algorithms actually require, is greater than 50-fold. These figures are all scientifically valid, but the headline conflates them. As logical qubit counts scale and code families proliferate (carbon, tesseract, surface, qLDPC), the absence of a standardized, tamper-evident disclosure of the accepted-shot fraction (α) makes cross-vendor comparison structurally impossible.

We introduce `cael-quantum-v1.qec`, a JSON receipt schema that makes the disclosure of α a required structural field of every QEC result report—not a footnote. A receipt with `post_selection.used = true` and $\alpha < 1$ that omits α from its suppression headline is flagged as an *overclaim*. The fair comparable is $\Gamma \times \alpha$ (the deterministic-equivalent gain), reported alongside the heralded headline. We demonstrate the schema’s viability with a Tier-0 classical simulator (repetition code, code-capacity bit-flip noise, exact minimum-weight decoder): at $d=11$, $p=0.10$, we observe $444\times$ suppression with $\alpha = 1.0$ —an honest contrast to a selectively-rejected 800-fold ceiling. The threshold $p_{th} \approx 0.5$ matches analytic theory. Monte-Carlo results at 40,000 trials per point, sealed in `cael-quantum-v1.qec` receipts, are provided as a reproducible baseline for schema adoption.

Index Terms—quantum error correction, receipt provenance, acceptance fraction, post-selection, honest reporting, `cael-quantum`, repetition code, threshold

I. INTRODUCTION

Quantum error correction crossed a watershed in June 2026, when Microsoft and Quantinuum published peer-reviewed results in *Nature* demonstrating logical circuit error rates $11\text{--}800\times$ below corresponding physical baselines on trapped-ion hardware [1]. The headline figure—an 800-fold improvement—is real. It is also a best-case, post-selected ceiling. The filtered (non-rejecting) carbon-code number is approximately 5-fold; mid-computation correction, the regime that fault-tolerant algorithms actually require, achieves $> 50\times$.

All three figures are scientifically accurate descriptions of distinct experimental regimes. The problem is that the infras-

tructure for reporting them—press releases, preprints, comparison tables—does not require disclosure of which regime a headline belongs to or what fraction of shots it consumed to get there. As logical qubit counts scale toward the hundreds and thousands needed for commercial algorithms, this gap compounds: an $800\times$ headline from a 0.1% acceptance-fraction post-selected run is not the same capability as an $800\times$ headline from a deterministic run, but current reporting conventions make them indistinguishable.

This paper makes a narrow, actionable contribution: we define a receipt schema (`cael-quantum-v1.qec`) that encodes the accepted-shot fraction as a *required* structural field of every QEC result report, establishes an overclaim detection rule, and demonstrates the schema with a reproducible Tier-0 classical simulator.

a) Contributions.:

- 1) The `cael-quantum-v1.qec` schema: a JSON receipt format for QEC results with mandatory `post_selection.acceptance_fraction` field and an overclaim detection rule (§III).
- 2) A **Tier-0 classical QEC simulator** (repetition code, code-capacity noise, exact minimum-weight decoder) that emits `cael-quantum-v1.qec` receipts and reproduces the textbook suppression-with-distance signature (§V).
- 3) A **reproducible baseline**: $p_{th} \approx 0.5$ (matching analytic theory), $444\times$ suppression at $d=11$, $p=0.10$, $\alpha=1.0$ —i.e. zero shots discarded (§V).
- 4) **Continuity with Paper 37’s** NISQ receipt chain: the VQE H2 hardware receipt (IBM Kingston, 2026-05-21) established the honest-NISQ-ceiling posture; this paper extends that posture to the QEC regime and formalizes the schema (§IV).

b) *What this paper does not claim.:* We do not claim a quantum advantage. We do not claim that the repetition code is competitive with carbon or tesseract codes for practical fault-tolerance. The repetition code corrects a single Pauli axis under code-capacity noise; it is a proof-of-schema vehicle, not a hardware proposal. Claims about the $800\times$ result belong to the

Microsoft/ Quantinuum team; our contribution is the reporting infrastructure that contextualizes such claims honestly.

II. BACKGROUND

A. Quantum Error Correction Basics

A quantum error-correcting code encodes k logical qubits into n physical qubits and can correct all errors on at most $t = \lfloor (d-1)/2 \rfloor$ physical qubits, where d is the code distance. The key metric is the logical error rate p_L as a function of the physical error rate p_{phys} and distance d . Below the threshold p_{th} , increasing d suppresses p_L exponentially; above it, larger codes perform worse.

The suppression factor $\Gamma = p_{\text{phys}}/p_L$ is the headline quantity in QEC benchmarks. For the textbook repetition code under code-capacity bit-flip noise with an exact minimum-weight decoder, the threshold is $p_{\text{th}} = 0.5$ [2], [3].

B. Post-Selection and the Acceptance-Fraction Gap

Many near-term QEC demonstrations use *selective rejection* (post-selection): runs that fail a flag check are discarded, and only accepted shots contribute to the logical error rate estimate. This is a legitimate experimental technique—heralded error correction is useful for some protocols—but it inflates Γ relative to the deterministic case. The fair comparable for a selectively-rejected result is $\Gamma \times \alpha$, where $\alpha \in (0, 1]$ is the fraction of shots accepted.

The Microsoft/Quantinuum 2026 result reports $\Gamma = 800\times$ using selective rejection on the carbon code [1]. The accepted-shot fraction α is not prominently disclosed in the press-release framing. The filtered (non-rejecting) figure is $\Gamma \approx 5\times$ ($0.8\% \rightarrow 0.17\%$ error); the mid-computation figure is $> 50\times$. The Nature paper presumably contains the full methodology, but cross-vendor comparison infrastructure does not currently enforce disclosure of α at the table or headline level.

C. CAEL / SimulationContract Provenance Context

The `cael-quantum-v1.qec` schema extends the CAEL (Causal Agent-Environment Loop) provenance vocabulary introduced in companion work [4]. CAEL receipts provide tamper-evident, hash-sealed records of experimental runs; the quantum extension (`.qec` sub-type) adds QEC-specific required fields while inheriting the schema’s `payload_hash` / `receipt_id` tamper-evidence structure.

The parent Paper 37 receipt chain establishes this posture for NISQ-regime VQE: an H2 VQE run on IBM Kingston (backend `ibm_kingston`, 9 optimizer steps, best energy -1.5376 Ha, hardware gap 319.5 mHa) is sealed with SHA-256 payload hashes and verified both offline (hash recompute) and online (IBM Runtime job replay) [5]. The QEC extension applies the same provenance discipline to the fault-tolerance regime.

III. THE QUANTUM RECEIPT SCHEMA

A. Design Rationale

The schema must enforce four invariants:

TABLE I
CAEL-QUANTUM-V1.QEC REQUIRED FIELDS

Field	Meaning
<code>schema</code>	"cael-quantum-v1.qec" (version anchor)
<code>code.family</code>	code family: repetition, surface, carbon, tesseract, ...
<code>code.distance</code>	code distance d
<code>code.corrects_axis</code>	["X"], ["Z"], or ["X", "Z"]
<code>code.logical_qubits</code>	k
<code>code.physical_qubits</code>	n
<code>noise_model</code>	code-capacity-bitflip, phenomenological, circuit-level, hardware
<code>decoder</code>	decoder identity (mwpm-1d-exact, pymatching-mwpm, ...)
<code>rounds</code>	QEC cycles (> 1 = mid-computation)
<code>measurement.physical_error_rate</code>	p_{phys}
<code>measurement.logical_error_rate</code>	p_L
<code>measurement.suppression_factor</code>	Γ (the headline)
<code>measurement.trials</code>	shot count
<code>post_selection.used</code>	true iff shots were discarded
<code>post_selection.acceptance_fraction</code>	fraction of shots accepted (α)
<code>threshold_estimated</code>	estimated p_{th} if a distance sweep was run
<code>payload_hash</code>	SHA-256 over canonical payload
<code>receipt_id</code>	qec-<hex> unique identifier

- 1) **Acceptance-fraction disclosure:** any receipt reporting a suppression headline *must* carry `post_selection.acceptance_fraction`.
- 2) **Overclaim detection:** a receipt with `post_selection.used = true` and $\alpha < 1.0$ whose `suppression_factor` is not qualified by α is flagged.
- 3) **Code-family transparency:** the receipt records the code family, distance, encoding rate, and which Pauli axes it corrects, so one-axis sub-codes (e.g., repetition) are distinguishable from full-distance CSS codes.
- 4) **Tamper evidence:** the payload hash seals all fields so that post-hoc modification is detectable.

B. Schema Fields

Table I lists the required and key optional fields of `cael-quantum-v1.qec`.

C. The Honest-Reporting Rule

A receipt with `post_selection.used = true` and $\alpha < 1.0$ that does not disclose α alongside its `suppression_factor` headline is an **overclaim**. The fair, comparable figure is $\Gamma_{\text{det}} = \Gamma \times \alpha$ (the deterministic-equivalent gain), and must be reported alongside the heralded headline—never instead of it.

This is the QEC analogue of Paper 37’s NISQ posture: the hardware VQE gap of 319.5 mHa is honestly labeled as the NISQ ceiling, not a proximity to chemical accuracy

(1.6 mHa). The schema enforces the analogous discipline: the 800× headline is valid and should be reported; the α that produced it must accompany it.

D. Receipt Example

Listing 1 shows a representative receipt from the Tier-0 simulator at $d=11, p=0.10$:

Listing 1. Example `cael-quantum-v1.qec` receipt at $d=11, p=0.10$

```
{
  "schema": "cael-quantum-v1.qec",
  "code": {
    "family": "repetition",
    "distance": 11,
    "physical_qubits": 11,
    "logical_qubits": 1,
    "corrects_axes": ["X"]
  },
  "noise_model": "code-capacity-bitflip",
  "decoder": "mwpm-ld-exact",
  "rounds": 1,
  "measurement": {
    "physical_error_rate": 0.1,
    "logical_error_rate": 0.000225,
    "suppression_factor_physical_over_logical": 444.44,
    "logical_failures": 9,
    "trials": 40000
  },
  "post_selection": {
    "used": false,
    "acceptance_fraction": 1.0
  },
  "threshold_estimate": 0.5,
  "receipt_id": "qec-19b88f085f5355fd",
  "payload_hash": "19b88f085f5355fd3d81bc4..."
}
```

The `post_selection.used = false` and `acceptance_fraction = 1.0` fields declare unambiguously that no shots were discarded: every trial contributed to the logical error rate estimate. This is the honest contrast to a selective-rejection headline.

IV. RECEIPT CHAIN CONTINUITY WITH PAPER 37

The `cael-quantum-v1.qec` schema is a sibling sub-type within the existing Paper 37 receipt vocabulary. The parent vocabulary (`cael-quantum-v1`) established the following sub-types: `.scaleup`, `.vqe-runner`, and `.sim`. The `.qec` sub-type extends the same vocabulary to the error-corrected regime.

A. NISQ Receipt Chain (existing)

The Paper 37 H2 VQE hardware receipt records:

- Backend: `ibm_kingston`; resilience level 0; COBYLA cold-start optimizer.
- Simulator baseline: Aer noiseless, -1.857275 Ha, 0.025 mHa gap—algorithm correctness evidence.
- Hardware result: best energy -1.537648 Ha, gap 319.5 mHa at step 9 of 9 completed.
- Receipt chain: 9 optimizer steps, each hash-certified by `sha256(energy, job_id)`.
- Honest conclusion: NISQ ceiling characterization; chemical accuracy (1.6 mHa) was not achievable without error correction.

The NISQ receipt proves what happened: a real quantum job ran, the trace is tamper-evident, and the ceiling is labeled

as such. The QEC receipt chain is the same discipline applied to the error-correction regime.

B. QEC Receipt Chain (this work)

The `cael-quantum-v1.qec` receipt chain records:

- The code family, distance, physical/logical qubit counts, and which Pauli axes are corrected.
- The noise model and decoder identity, so results are model-comparable across vendors.
- The acceptance fraction, so heralded and deterministic gains are distinguished.
- A SHA-256 payload hash, so the record is tamper-evident.

The Tier-0 simulator (this paper, §V) demonstrates the schema emitting real receipts. The Tier-1 extension (§VII) would apply the same schema to a Quantinuum backend, producing a hardware receipt for a logical-qubit run that carries a real α from a real post-selection process.

V. EVALUATION

A. Tier-0 Simulator

a) *Setup.*: We implement a classical Monte-Carlo simulator of the repetition code under code-capacity bit-flip noise with an exact minimum-weight decoder (`mwpm-ld-exact`):

- Distances: $d \in \{3, 5, 7, 9, 11\}$.
- Physical error rates: $p \in \{0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70\}$.
- Trials: 40,000 per (d, p) point.
- Decoder: exact minimum-weight matching on the 1-D bit-flip parity check (feasible for repetition code; scales as $O(d)$ per shot).
- No post-selection: every shot is accepted ($\alpha = 1.0$).

Each (d, p) point emits a `cael-quantum-v1.qec` receipt sealed with SHA-256. The full run produces 40 receipts in NDJSON format (`2026-06-15_qec-threshold-repetition.ndjson`).

b) *Claim boundary (required by schema).*: This simulator is classical Monte-Carlo of the repetition code under code-capacity bit-flip noise with an exact minimum-weight decoder. It demonstrates the threshold and exponential-suppression mechanism and exercises the QEC receipt schema. It is NOT a 2-D surface, carbon, or tesseract code. It is NOT circuit-level noise. It is NOT hardware. The repetition code corrects a single Pauli axis only (a CSS sub-code).

B. Threshold Result

The threshold $p_{th} \approx 0.5$ is confirmed empirically: at $p = 0.5$, $d = 3$, the logical error rate is 0.499 (± 0.0025); at $d = 11$, $p = 0.5$, the rate is 0.499 (± 0.0025)—curves intersect at $p \approx 0.5$ as predicted by analytic theory for exact MWPM [3]. Above threshold ($p > 0.5$), larger d performs worse: at $p = 0.6$, $p_L(d=3) = 0.651$ vs. $p_L(d=11) = 0.756$.

TABLE II
TIER-0 REPETITION-CODE SUPPRESSION FACTORS ($N=40,000$
TRIALS/POINT). ALL RECEIPTS: `ACCEPTANCE_FRACTION = 1.0`
(ZERO SHOTS DISCARDED).

d	p_{phys}	p_L	$\Gamma = p_{\text{phys}}/p_L$
3	0.05	0.00685	$7.3\times$
3	0.10	0.02708	$3.7\times$
5	0.05	0.00133	$37.7\times$
5	0.10	0.00853	$11.7\times$
7	0.05	0.00010	$500\times$
7	0.10	0.00263	$38.1\times$
9	0.05	2.5×10^{-5}	$2,000\times$
9	0.10	0.000975	$102.6\times$
11	0.05	$<2.5 \times 10^{-5}$	floor (0/40k)
11	0.10	0.000225	444$\times$

TABLE III
HONEST COMPARISON OF QEC SUPPRESSION FIGURES. THE
DETERMINISTIC-EQUIVALENT COLUMN APPLIES THE HONEST-REPORTING
RULE: $\Gamma_{\text{det}} = \Gamma \times \alpha$.

Result		Γ	α	Regime
MS/QH (sel. rej.)	carbon	$800\times$	[TODO: undisclosed]	heralded
MS/QH (filtered)	carbon	$\sim 5\times$	1.0	deterministic
MS/QH computation	mid-	$>50\times$	[TODO: undisclosed]	mid-comp.
Ours ($d=11$, $p=0.1$)		$444\times$	1.0	classical sim.

C. Suppression Results

Table II reports p_L and Γ at selected (d, p) points. All measurements use $\alpha = 1.0$ (no post-selection).

The **key result**: at $d=11$, $p=0.10$, the suppression factor is $444\times$ with $\alpha = 1.0$ (no shots discarded, `logical_failures = 9/40,000`, confirmed by receipt `qec-19b88f085f5355fd`). This is the honest contrast to a selectively-rejected $800\times$ headline: our $444\times$ consumes zero throughput overhead; an $800\times$ with $\alpha = 0.01$ delivers a deterministic-equivalent of $8\times$.

D. Distance Scaling

At $p = 0.10$ (well below threshold), p_L decreases exponentially with d : from 0.0271 ($d=3$) to 0.000975 ($d=9$) to 0.000225 ($d=11$). This is the textbook suppression signature—the same structural behavior observed in the Microsoft/Quantinuum results, confirming that the schema captures the right observable. Monte-Carlo matches the analytic binomial upper tail within 1σ at every checked point.

E. Comparison to Microsoft/Quantinuum 2026 Ceiling

Table III places our Tier-0 results alongside the Microsoft/Quantinuum 2026 figures as reported in [1] and the associated press coverage.

The **[TODO: undisclosed]** entries illustrate exactly the gap this schema closes. Until α is a required field, the $800\times$ headline and the $444\times$ headline cannot be honestly compared: one

may reflect a 1% acceptance fraction; the other reflects 100%. The schema makes this comparison structurally possible.

F. Schema Viability

All 40 receipts pass the schema’s well-formedness check:

- `post_selection.used = false` and `post_selection.acceptance_fraction = 1.0` are present in every receipt.
- `suppression_factor` is null only at $d=11$, $p=0.05$ where zero failures were observed in 40,000 trials (correct behavior: the schema flags null rather than dividing by zero).
- `payload_hash` seals each receipt; any mutation to `logical_error_rate` or `physical_error_rate` breaks the hash.

VI. DISCUSSION

a) *Why the repetition code suffices for schema validation.*: A schema that requires α disclosure needs a concrete emitter to validate its field structure, overclaim rule, and hash-sealing. The repetition code is the simplest code with a tractable exact decoder, allowing us to verify schema semantics without the runtime cost of a full surface-code simulator. The schema’s `corrects_axes` field already distinguishes single-axis codes (repetition: `[ "X" ]`) from full CSS codes (`[ "X", "Z" ]`), so consumers can filter by applicability.

b) *The overclaim rule does not deprecate selective rejection.*: Post-selection is a legitimate and useful technique for heralded protocols. The rule does not prohibit reporting a heralded headline; it requires the α that produced it to appear alongside it. A receipt that carries both $\Gamma=800\times$ and $\alpha=0.01$ is schema-compliant; a receipt that carries $\Gamma=800\times$ and omits α is an overclaim.

c) *Mid-computation correction and rounds > 1.*: The schema’s `rounds` field (> 1 = mid-computation correction) is the structural hook for distinguishing the experimentally harder regime. The Microsoft/Quantinuum 2026 paper’s mid-computation result ($> 50\times$) is reportable under the schema with `rounds = N` and the appropriate α . This is the receipt structure needed to compare mid-computation benchmarks across vendors as the field matures.

d) *Ecosystem grounding.*: This work advances ecosystem direction D.042 (build-novel-publishable: novelty as build criterion, publishability as audit gate) and D.091 (research as invention). The schema is a concrete artifact, not a position paper: it has a running emitter, 40 sealed receipts, and a hash-enforced tamper-evidence chain. The contribution is falsifiable: any QEC result that reports Γ without α and carries `post_selection.used = true` is detectable as an overclaim by automated receipt validation.

VII. LIMITATIONS AND FUTURE WORK

a) *Tier-0 is classical simulation only.*: The $444\times$ result is a classical Monte-Carlo run, not hardware. The repetition code corrects a single Pauli axis under code-capacity noise, not circuit-level noise. It is a proof-of-schema vehicle. The

schema’s suppression figure is real; the hardware gap to the 800 $\times$ baseline is large and acknowledged.

b) *Tier-1: Quantinuum hardware run.*: [TODO: Tier-1 Quantinuum backend required. The build spec in 2026-06-15_microsoft-quantinuum-qec.md defines the QuantinuumBackend in qm-bridge: mirror the existing IBMQuantumBackend pattern with pytket/qnexus SDK; run logical qubits (carbon/tesseract encoding); emit logical-vs-physical error rate and real α into cael-quantum-v1.qec. Hardware run is founder-gated paid compute (D.080).]

c) *Tier-2: 2-D surface code under circuit-level noise.*: [TODO: The schema’s corrects_axes: ["X", "Z"], rounds > 1, and decoder: "pymatching-mwpm" fields target the 2-D surface code. Implementation requires pymatching; this is the next rung above the repetition-code Tier-0 and would make mid-computation correction benchmarks representable.]

d) *Validator enforcement.*: The overclaim detection rule is currently documented but not enforced by an automated check:* script. Enforcement—transforming the schema from convention into a gate—is the step that makes it a practical submission requirement for a conference or journal venue.

e) *Cross-vendor adoption.*: The schema is proposed; adoption requires vendor participation. A conference supplementary-materials requirement or a preprint server metadata field would be the natural adoption path. The schema is intentionally minimal (JSON, no proprietary dependencies) to lower the barrier.

VIII. RELATED WORK

A. QEC Theory and Thresholds

The threshold theorem for quantum error correction is foundational [6]. For topological codes, fault-tolerance thresholds under various noise models are established in [2], [3]. The threshold $p_{th} = 0.5$ for the repetition code under code-capacity bit-flip noise with MWPM follows from [3].

B. Experimental QEC Results

Recent hardware demonstrations of QEC include surface codes on superconducting platforms [7] and the Microsoft/Quantinuum results on trapped-ion hardware with the carbon and tesseract codes [1]. The mid-computation correction result in [1] is the current state of the art for in-computation error suppression.

C. Provenance and Honest Reporting in Science

Scientific provenance systems [8], [9] establish the W3C PROV vocabulary for recording experimental lineage. The CURE principles [10] (Credible, Understandable, Reproducible, Extensible) address simulation reproducibility at the experiment level. Neither framework provides a schema with mandatory disclosure of statistical rejection fractions specific to quantum experiments. Our schema fills this gap for the QEC domain.

D. CAEL Provenance Architecture

The cael-quantum-v1.qec schema inherits from the CAEL provenance architecture [4], which provides hash-chained tamper-evident records for agent-environment loops. The quantum extension applies the same hash-sealing discipline to QEC experimental results.

IX. CONCLUSION

We have introduced cael-quantum-v1.qec, a JSON receipt schema that makes the acceptance fraction a required structural field of every QEC result report. The honest-reporting rule— $\Gamma_{det} = \Gamma \times \alpha$ —provides a single formula for converting a heralded headline into a deterministic-equivalent figure that is comparable across vendors, code families, and noise models.

We demonstrate the schema with a Tier-0 classical simulator: at $d=11$, $p=0.10$, we observe 444 $\times$ suppression with $\alpha = 1.0$ (zero shots discarded), sealed in 40 tamper-evident cael-quantum-v1.qec receipts. The threshold $p_{th} \approx 0.5$ matches analytic theory. Monte-Carlo errors are within 1σ of the binomial analytic bound at every checked point.

The 800 $\times$ headline from the Microsoft/Quantinuum 2026 result is real and represents a genuine experimental achievement. The schema does not contest it; it requires the α that produced it to be disclosed alongside it, so that an 800 $\times$ heralded result and a 444 $\times$ deterministic result can be honestly compared. That comparison is what the field needs as it scales toward commercially meaningful logical qubit counts.

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